

Laboratory Work 1

ERD Diagram — International Airport Database

Course: Databases

Student: _____

Group: _____

1. Introduction

The International Airport Database is designed to manage airport operations, including airports, airlines, flights, passengers, bookings, boarding passes, baggage and security checks. The database uses primary and foreign keys to maintain data integrity and represent relationships between entities.

2. Entities and Attributes

Airport		
airport_id	INT	PK
airport_name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	NULL
city	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Airline		
airline_id	INT	PK
airline_code	VARCHAR(10)	PK? / UNIQUE
name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Flight		
flight_id	INT	PK
departing_gate	VARCHAR(10)	NULL
arriving_gate	VARCHAR(10)	NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
airline_id	INT	FK
departure_airport_id	INT	FK
arrival_airport_id	INT	FK
scheduled_departure_time	TIMESTAMP	NOT NULL
scheduled_arrival_time	TIMESTAMP	NOT NULL
actual_departure_time	TIMESTAMP	NULL
actual_arrival_time	TIMESTAMP	NULL

Passenger		
passenger_id	INT	PK
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
gender	VARCHAR(20)	NOT NULL
date_of_birth	DATE	NOT NULL
country_of_citizenship	VARCHAR(100)	NOT NULL
country_of_residence	VARCHAR(100)	NOT NULL
passport_number	VARCHAR(30)	UNIQUE, NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Booking		
booking_id	INT	PK
flight_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL
status	VARCHAR(30)	NOT NULL
booking_platform	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
ticket_price	DECIMAL(10,2)	NOT NULL

BookingChange		
booking_change_id	INT	PK
booking_id	INT	FK, NOT NULL
change_description	TEXT	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

BoardingPass		
boarding_pass_id	INT	PK
seat	VARCHAR(10)	NOT NULL
boarding_time	TIMESTAMP	NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
booking_id	INT	FK, NOT NULL

Baggage		
baggage_id	INT	PK
weight_kg	DECIMAL(6,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
booking_id	INT	FK, NOT NULL

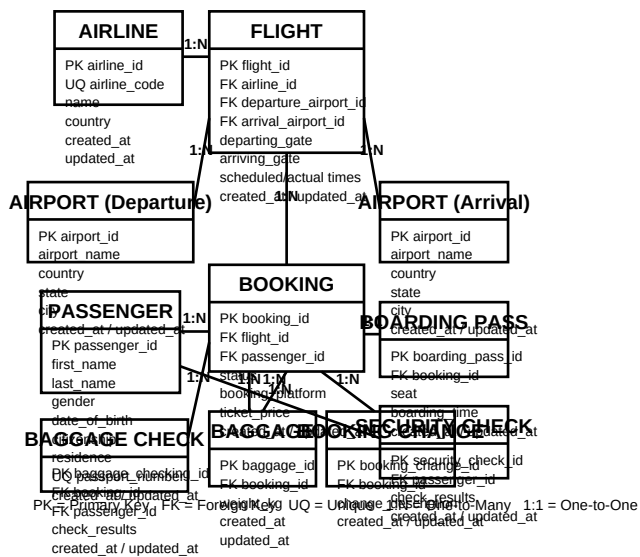
BaggageCheck		
baggage_checking_id	INT	PK
check_results	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
booking_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL

SecurityCheck		
security_check_id	INT	PK
check_results	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
passenger_id	INT	FK, NOT NULL

3. Relationships and Constraints

Entity A	Entity B	Cardinality	Description
Airline	Flight	1 : N	One airline operates many flights; each flight belongs to one airline.
Airport	Flight (departure)	1 : N	One airport can be the departure airport for many flights.
Airport	Flight (arrival)	1 : N	One airport can be the arrival airport for many flights.
Passenger	Booking	1 : N	One passenger can have many bookings; each booking belongs to one passenger.
Flight	Booking	1 : N	One flight can have many bookings; each booking belongs to one flight.
Booking	BoardingPass	1 : 1	In this simplified model, one booking has one boarding pass.
Booking	Baggage	1 : N	One booking can contain multiple baggage records.
Booking	BookingChange	1 : N	One booking can have multiple recorded changes.
Booking	BaggageCheck	1 : N	A booking can have multiple baggage-check records.
Passenger	BaggageCheck	1 : N	A passenger can be associated with multiple baggage checks.
Passenger	SecurityCheck	1 : N	A passenger can have multiple security-check records.

4. ERD Diagram



5. Legend

Symbol / Type	Meaning
PK	Primary Key — uniquely identifies each record.
FK	Foreign Key — references a primary key in another entity.
UQ / UNIQUE	The value must be unique.
NOT NULL	The attribute is required.
1:N	One record can be related to many records.
1:1	One record is related to exactly one record in the simplified model.
INT	Integer number.
VARCHAR	Variable-length text.
DATE	Date value.
TIMESTAMP	Date and time.
DECIMAL	Number with decimal places.
TEXT	Long text information.

6. Conclusion

The ERD represents the main data structures and relationships required for the simplified International Airport Database. Primary keys uniquely identify records, while foreign keys connect related entities. The model supports flight scheduling, passenger bookings, boarding passes, baggage registration and checking, booking changes, and passenger security checks.